DPP No: 43

Maximum Time 50 Min

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C³

(COMPETISHUN CRASH COURSE)

PHYSICS

TARGET JEE-MAINS

SYLLABUS : STRING WAVE & SOUND WAVE

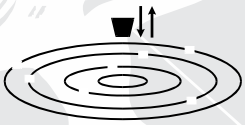
1. A boat at anchor is rocked by waves whose crests are 100m apart and velocity is 25m/s. The boat bounces up once in every :–

(A) 2500 s (B) 75 s (C) 4 s (D) 0·25 s

2. When sound wave is refracted from air to water, which of the following will remain unchanged?

(A) wave number (B) wavelength (C) wave velocity (D) frequency

3. A piece of cork is floating on water in a small tank. The cork oscillates up and down vertically when small ripples pass over the surface of water. The velocity of the ripples being 0.21 ms^{-1}, wave length 15 mm and amplitude 5 mm, the maximum velocity of the piece of cork is ($\pi = \frac{22}{7}$)



(A) 0.44 ms^{-1} (B) 0.24 ms^{-1} (C) 2.4 ms^{-1} (D) 4.4 ms^{-1}

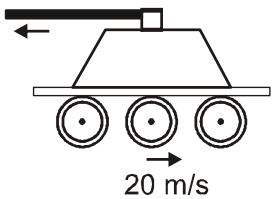
4. The frequency of a man's voice is 300 Hz and its wavelength is 1 meter. If the wavelength of a child's voice is 1.5 m, then the frequency of the child's voice is:

(A) 200 Hz (B) 150 Hz (C) 400 Hz (D) 350 Hz.

5. A wave of frequency 500 Hz travels between X and Y, a distance of 600 m in 2 sec. How many wavelength are there in distance XY:–

(A) 1000 (B) 300 (C) 180 (D) 2000

6. A machine gun is mounted on an armored car moving with a speed of 20 ms^{-1}. The gun can point against the direction of motion of car. The muzzle speed of bullet is equal to speed of sound in air i.e., 340 ms^{-1}. The time difference between bullet actually reaching and sound of firing reaching at a target 544 m away from car at the instant of firing is



(A) 1.2 s (B) 0.1 s (C) 1 s (D) 10 s

7. A tuning fork is vibrating with constant frequency and amplitude. If the air is heated without changing pressure the following quantities will increase.

(A) Wavelength (B) Frequency (C) Velocity (D) Time period

8. The ratio of speed of sound in a monoatomic gas to that in water vapours at any temperature is. (when molecular weight of gas is 40 gm/mol and for water vapours is 18 gm/mol)

(A) 0.75 (B) 0.73 (C) 0.68 (D) None of these

9. Under similar conditions of temperature and pressure, In which of the following gases the velocity of sound will be largest.

(A) H_2 (B) N_2 (C) He (D) CO_2

10. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of:–

(A) 1000 (B) 10000 (C) 10 (D) 100

11. A sound of intensity I is greater by 3.0103 dB from another sound of intensity 10 nW cm^{-2}. The absolute value of intensity of sound level I in Wm^{-2} is :

(A) 2.5×10^{-4} (B) 2×10^{-4} (C) 2.0×10^{-2} (D) 2.5×10^{-2}

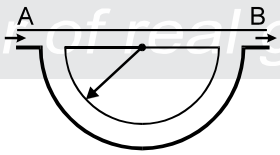
12. For a sound source of intensity I W/m^2, corresponding sound level is B_0 decibel. If the intensity is increased to 4I, new sound level becomes approximately :

(A) $2B_0$ dB (B) $(B_0 + 3)$dB (C) $(B_0 + 6)$dB (D) $4B_0$ dB

13. The sound intensity is 0.008 W/m^2 at a distance of 10 m from an isotropic point source of sound. The power of the source is approximately :

(A) 2.5 watt (B) 0.8 watt (C) 8 watt (D) 10 watt

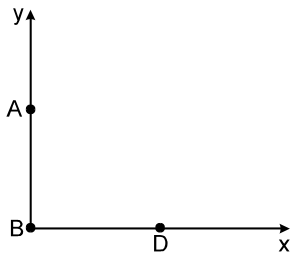
14. Sound signal is sent through a composite tube as shown in the figure. The radius of the semicircular portion of the tube is r. Speed of sound in air is v. The source of sound is capable of giving varied frequencies in the range of ν_1 and ν_2 (where $\nu_2 > \nu_1$). If n is an integer then frequency for maximum intensity is given by :



(A) $\frac{nv}{r}$ (B) $\frac{nv}{r(\pi-2)}$ (C) $\frac{nv}{\pi r}$ (D) $\frac{nv}{(r-2)\pi}$

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15. An interference is observed due to two coherent sources 'A' & 'B' separated by a distance 4λ along the y–axis where λ is the wavelength of the source. A detector D is moved on the positive x–axis. The number of points on the x–axis excluding the points, x = 0 & x = ∞ at which maximum will be observed is -



(A) three (B) four (C) two (D) infinite

16. When a sound wave is reflected from a wall, the phase difference between the reflected and incident pressure wave is:

(A) 0 (B) π (C) $\pi/2$ (D) $\pi/4$

17. A cylindrical tube, open at both ends, has a fundamental frequency f in air. The tube is dipped vertically in water so that half of its in water. The fundamental frequency of the air column is now:–

(A) $\frac{f}{2}$ (B) $\frac{3f}{4}$ (C) f (D) $2f$

18. At the closed end of an organ pipe :

(A) the displacement is zero (B) the displacement amplitude is maximum

(C) the pressure amplitude is zero (D) the pressure amplitude is maximum

19. The fundamental frequency of a closed organ pipe is same as the first overtone frequency of an open pipe. If the length of open pipe is 50 cm, the length of closed pipe is

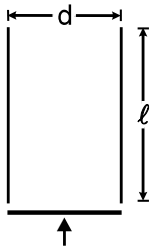
(A) 25 cm (B) 12.5 cm (C) 100 cm (D) 200 cm

20. An organ pipe P_1 closed at one end vibrating in its first harmonic and another pipe P_2 open at ends vibrating in its third harmonic are in resonance with a given tuning fork. The ratio of the length of P_1 and P_2 is:–

(A) $\frac{8}{3}$ (B) $\frac{3}{8}$ (C) $\frac{1}{6}$ (D) $\frac{1}{3}$

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21. A tube of diameter d and of length ℓ unit is open at both the ends. Its fundamental frequency of resonance is found to be ν_1. The velocity of sound in air is 330 m/sec. One end of tube is now closed. The lowest frequency of resonance of tube is now ν_2. Taking into consideration the end correction, $\frac{\nu_2}{\nu_1}$ is



(A) $\frac{(\ell+0.6d)}{(\ell+0.3d)}$ (B) $\frac{1}{2}\frac{(\ell+0.3d)}{(\ell+0.6d)}$ (C) $\frac{1}{2}\frac{(\ell+0.6d)}{(\ell+0.3d)}$ (D) $\frac{1}{2}\frac{(d+0.3\ell)}{(d+0.6\ell)}$

22. A resonance tube is resonated with tuning fork of frequency 256 Hz. If the length of first and second resonating air columns are 32 cm and 100 cm, then end correction will be

(A) 1 cm (B) 2 cm (C) 4 cm (D) 6 cm

23. Frequency of tuning fork A is 256 Hz. It produces 4 beats/second with tuning fork B. When wax is applied at tuning fork B then 6 beats/second are heard. Frequency of B is:–

(A) 250 Hz (B) 260 Hz

(C) 252 Hz (D) (A) & (C) both may possible

24. A sound source of frequency 512 Hz is producing 6 beats with a guitar. If the string of guitar is stretched slightly then beat frequency decreases. The original frequency of guitar is

(A) 506 Hz (B) 512 Hz (C) 518 Hz (D) 524 Hz

25. A tuning fork of known frequency 256 Hz. makes 5 betas per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per second when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was :

(A) (256 + 2) Hz. (B) (256 – 2) Hz (C) (256 – 5) Hz (D) (256 + 5) Hz

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ANSWER KEY

1.	(C)	**2.**	(D)	**3.**	(A)	**4.**	(A)	**5.**	(A)
6.	(B)	**7.**	(AC)	**8.**	(A)	**9.**	(A)	**10.**	(D)
11.	(B)	**12.**	(C)	**13.**	(D)	**14.**	(B)	**15.**	(A)
16.	(A)	**17.**	(C)	**18.**	(AD)	**19.**	(B)	**20.**	(C)
21.	(C)	**22.**	(B)	**23.**	(C)	**24.**	(A)	**25.**	(C)